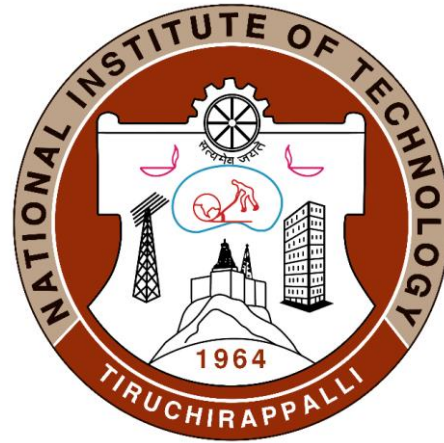


Introduction to Composites Material



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Metal Matrix composites

Functions of the Matrix Material (Primary Phase)

- It takes the load and transfer it to the reinforcement
- It binds or holds the reinforcement phase and protects the same from external damage (either chemical and mechanical)
- Matrix also separates the individual fibers and prevents brittle cracks from passing completely across the section of the composite

Historical Significance of MMCs

1. Initial work in late 1960s was stimulated by the high-performance needs of the aerospace industry
2. Steel wire reinforced copper, were among the first continuous-fiber reinforced metal composites
3. Boron filament was the first high strength, high modulus reinforcement in MMCs

Generally, metals with low density coupled with their good low temperature toughness are preferred for MMCs.

However, metals not only offer greater strength and stiffness but can also withstand much higher temperature than polymer matrices.

In addition, metals exhibited superior fracture toughness

Metal matrix composites (MMC) is composite material with at least two constituent parts, one being a metal necessarily, the other material may be a different metal or another material such as a ceramic or organic compound

When at least three materials are present, it is called a hybrid composite

The most used metal matrices are Al, Mg and Ti and their alloys which are well suited for aircraft structural applications owing to their low density

The value of elastic modulus of these metals is 69, 45 and 110 GPa respectively.

Why we use MMCs

1. High strength to density ratio
2. High stiffness to density ratio
3. Toughness
4. Good wear resistance
5. Damping capacity
6. Specific modulus
7. Lower coefficient of thermal expansion
8. Better abrasion resistance
9. Superior creep and fatigue properties

Typical fibers and matrixes for Metal Matrix composites and characteristics values of relevant properties

	Density (g/cm ³)	Thermal Expansion (10 ⁻⁶ K ⁻¹)	Stiffness (GPa)	Strength (MPa)
Fiber				
Glass	2.5	5	70	2500
Aramid	1.4	-6/+75	140	2500
Carbon/graphite	1.8	-1/+15	200-700	2500
Boron	2.6	5	400	7000
Al ₂ O ₃	4.0	7	390	2800
(Saffil) short	4	8	300	1500
(FP) long	4	8	370	2100
SiC	3.2	5	390	
(Whisker) short	2.6	4	300	1500
(Nicalon) long	2.6	4	200-600	3000-8000
W	19.3	4	400	4000
Matrix				
Mg	1.7	26	45	180
Al	2.7	23	70	35
Ti	4.5	9	120	170
Cu	8.9	17	130	100
Ni	8.9	13	220	120

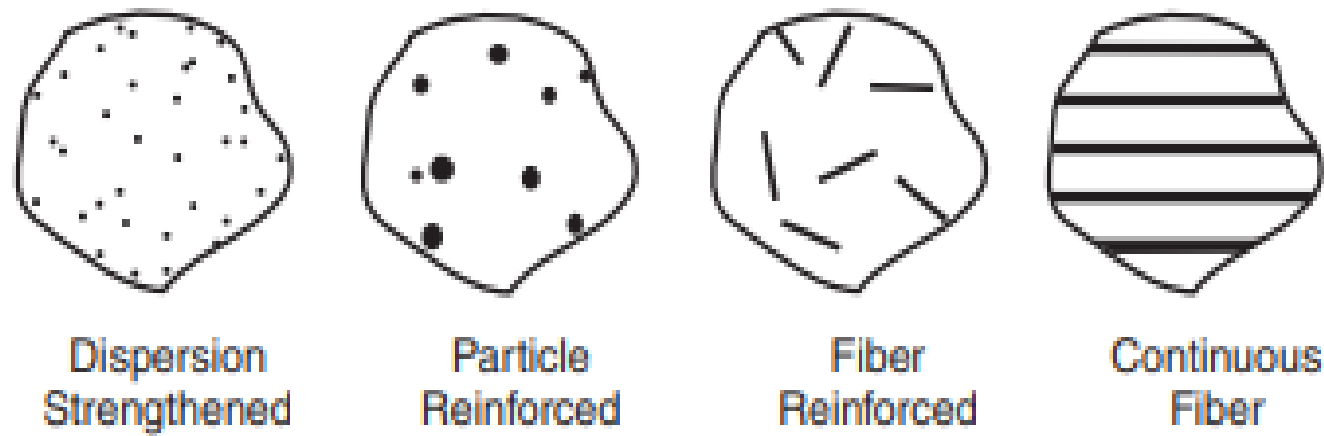


Fig. Different types of reinforcement used in metal-matrix composites

Applications of MMCs

Drive shaft

Material : $\text{AlMg1SiCu} + 20\% \text{Al}_2\text{O}_3\text{P}$

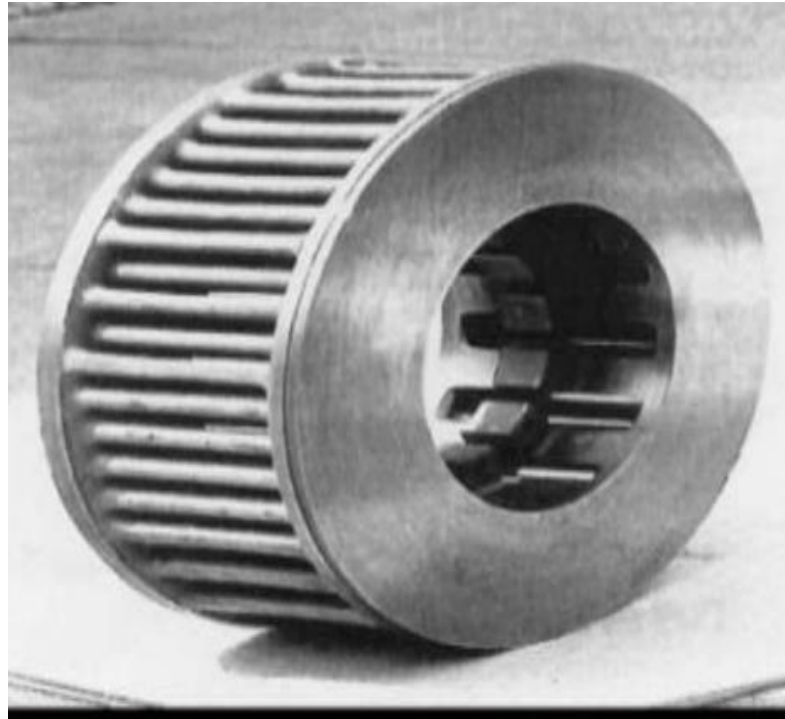
Processing : Extrusion from cast feed material



Vented Passenger car brake disk

Material : G-AlSi12Mg+20%Vol SiC

Processing : Sand/Die casting



F-16 Fighting Falcon uses monofilament silicon carbide fibers in a Ti matrix for a structural components of the jets landing gear



Processing of MMCs

1. Solid state methods
2. Semi-solid methods
3. Liquid state methods
4. Vapor deposition
5. In-situ fabrication Techniques

Processing of Metal Matrix composites



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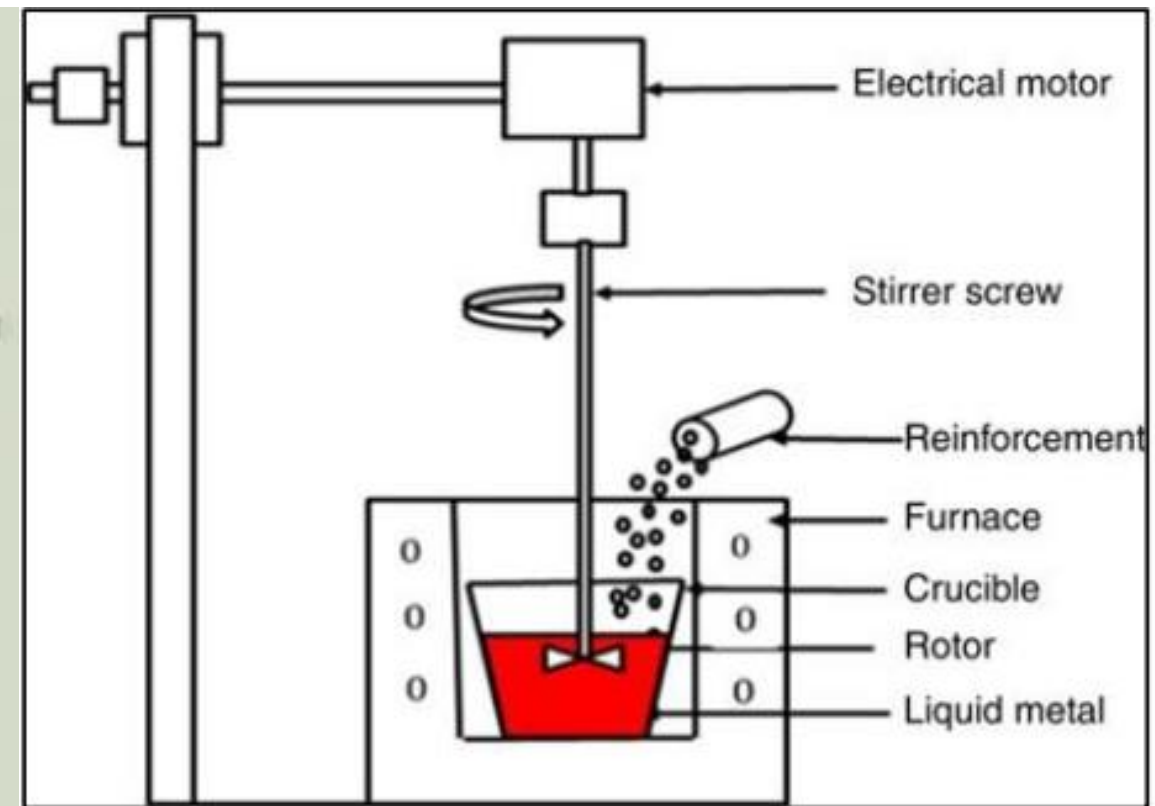
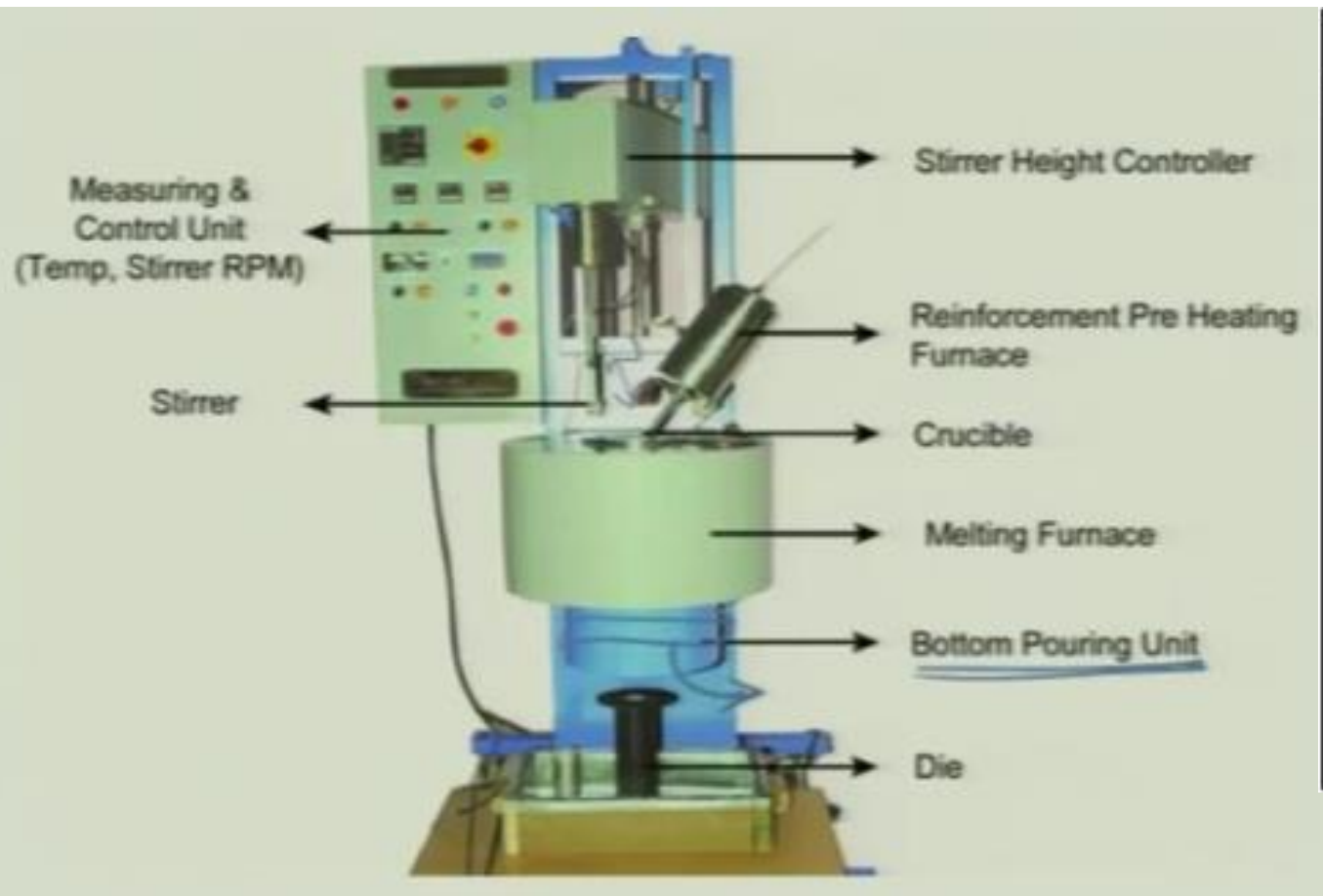
Liquid state :

- It involves incorporation of dispersed phase into a molten matrix metal.
- In order to provide high level of mechanical properties of the composite, good interfacial bonding between the dispersed phase and the liquid matrix is obtained.
- This can be achieved by coating the dispersed phase particles.
- It also prevents the chemical interaction between the dispersed phase and the matrix.

Stir casting process:

It is a liquid state method of composite material fabrication, in which a dispersed phase is mixed with a molten matrix metal by means of mechanical stirring.

It is the simplest and the most cost effective method of liquid state fabrication.



Some features of Stir casting process:

- **Content of dispersed phase is limited (usually not more than 30 Vol.%)**
- **Distribution of dispersed phase throughout the matrix is not perfectly homogeneous**

Clusters of the dispersed particles

Gravity segregation (due to difference in densities of the dispersed and matrix phase)

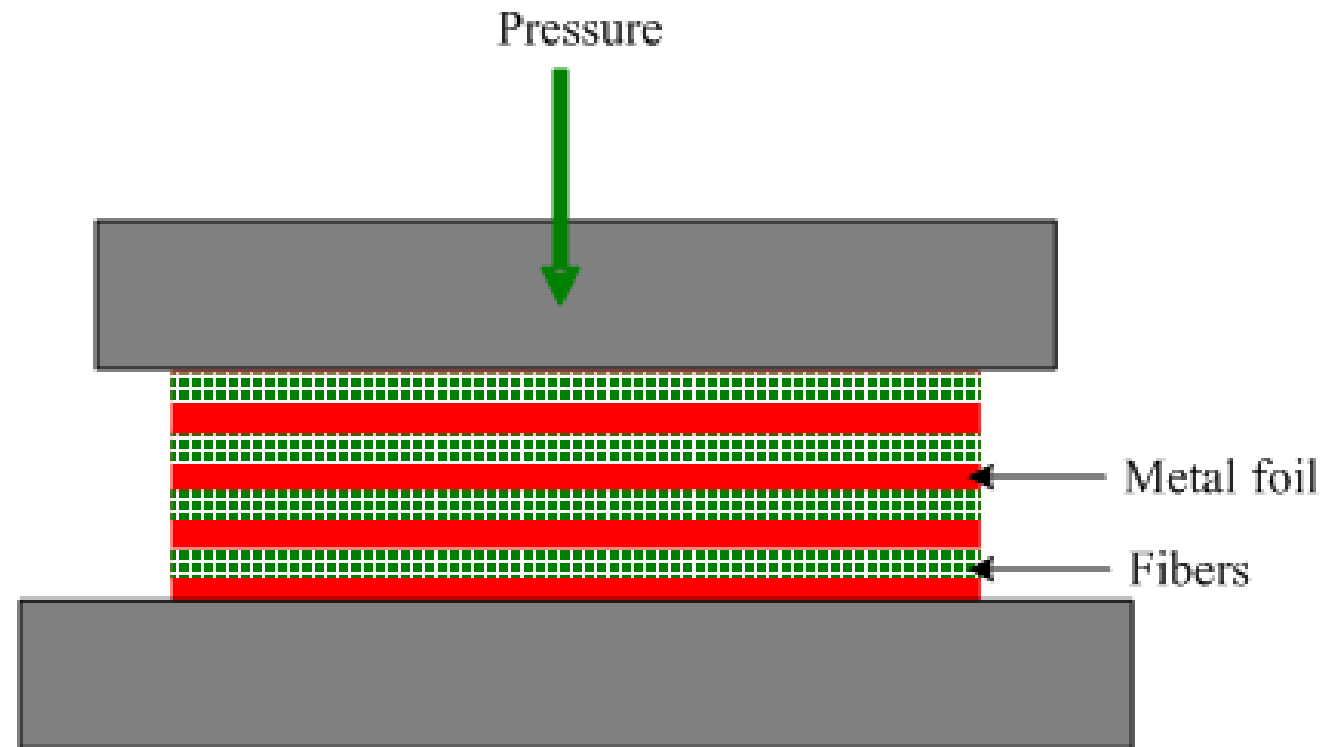
Solid state:

- It is the process in which MMC are formed as a result of bonding matrix metal and dispersed phase due to mutual diffusion, occurring between them in solid states at elevated temperature and under pressure.

Diffusion bonding

- Diffusion bonding is a solid state fabrication method, in which a matrix in form of foils and a dispersed phase in form of long fibers are stacked in a particular order and then pressed at elevated temperature.
- Multilayer structure (laminate composite)
- Used for fabrication of simple shape parts. (plates, tubes)

Diffusion Bonding



Thank you